Flexibility Management Needs Privacy-by-Design

How Can You Use Your Flexibility Without Sharing Your Smart Meter Data?

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Lecture 11a

“This research was funded by the CETPartnership, under the Joint Call 2023. The CETPartnership’s research projects are co-funded by the European Commission (Grant Agreement No. 101069750) and the national funding organizations listed on the CETPartnership website, Project No. CETP-2023-00493.”

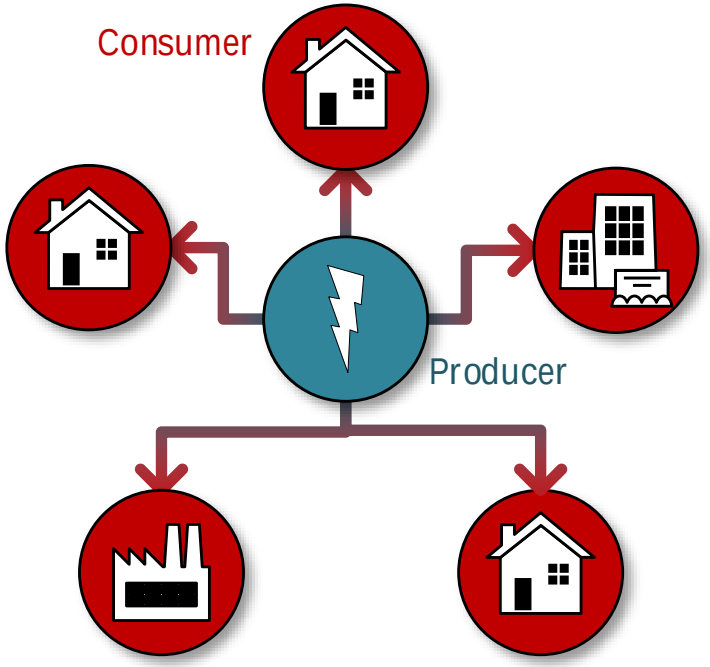
Changing the approach: The smart grid

Traditional Power Grid

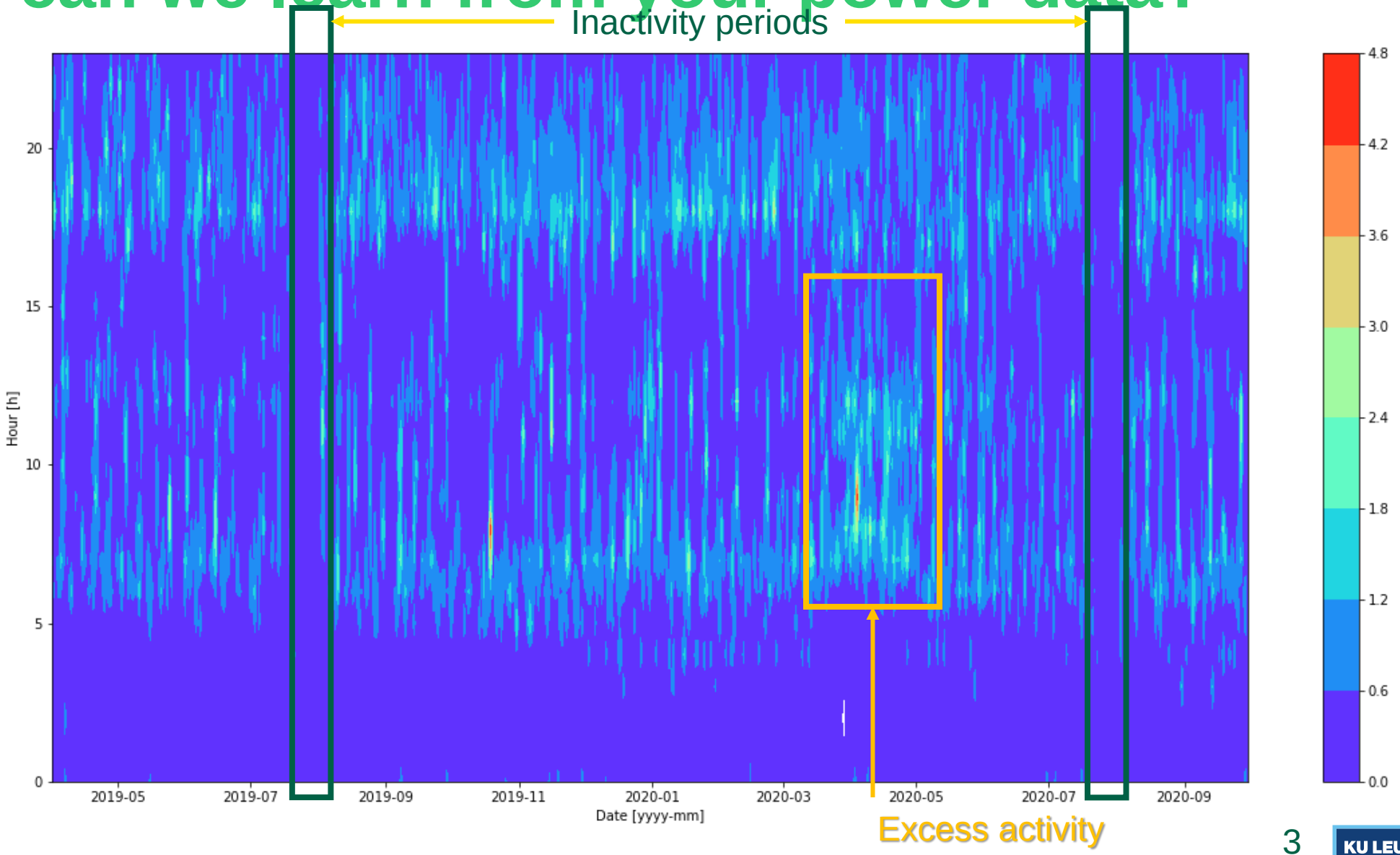
- Centralised: Producer → Consumer
- Consumers = Passive agents
  - Poor dispatchability of renewables sources
  - Heterogeneous and uncertain behaviour

Smart Grid

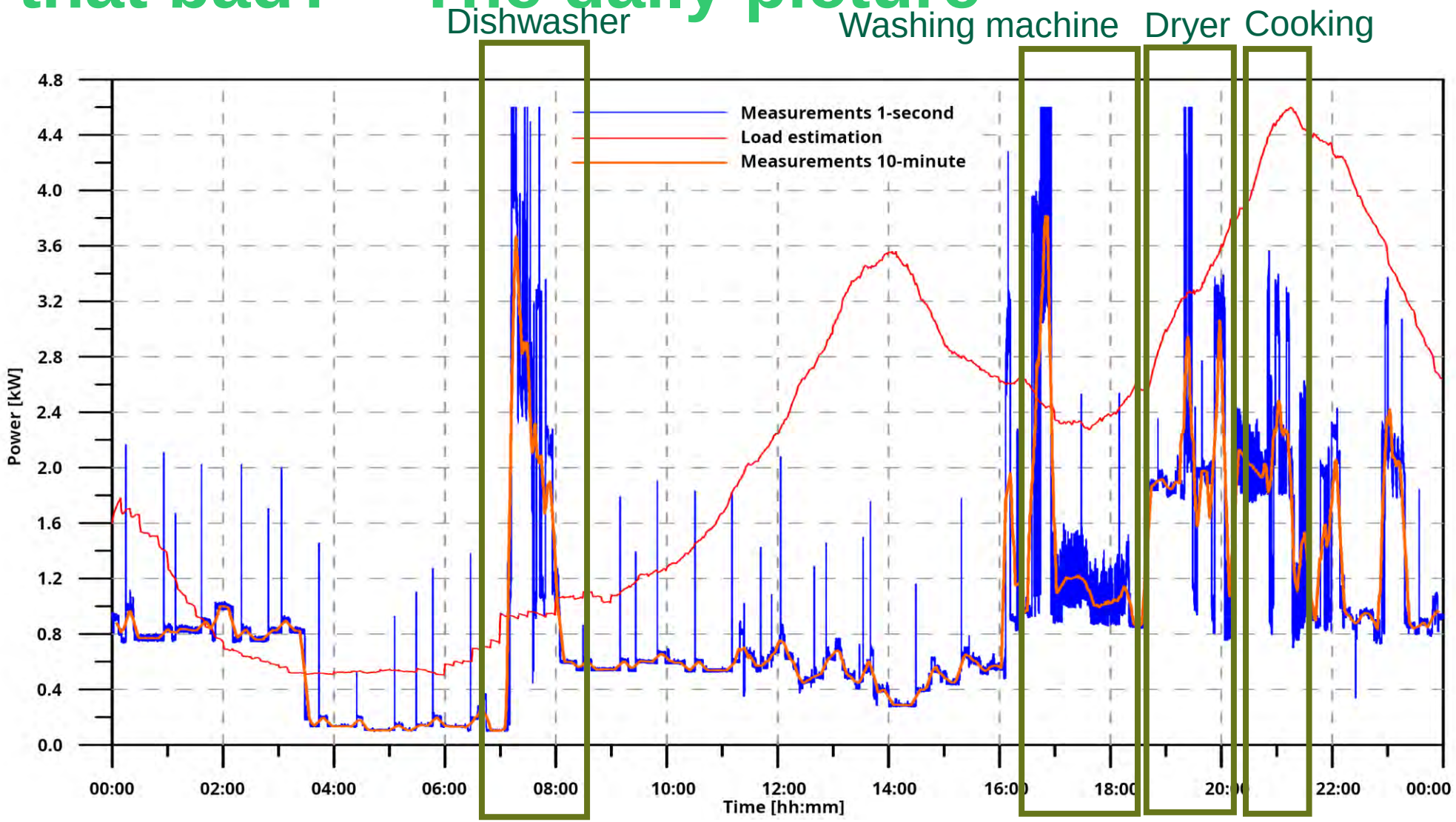
- Distributed: Producer ↔ Consumer
- Prosumers = Active agents
  - Bi-direction power flow; Feed-in tariffs
  - Demand response, Energy Management



What can we learn from your power data?

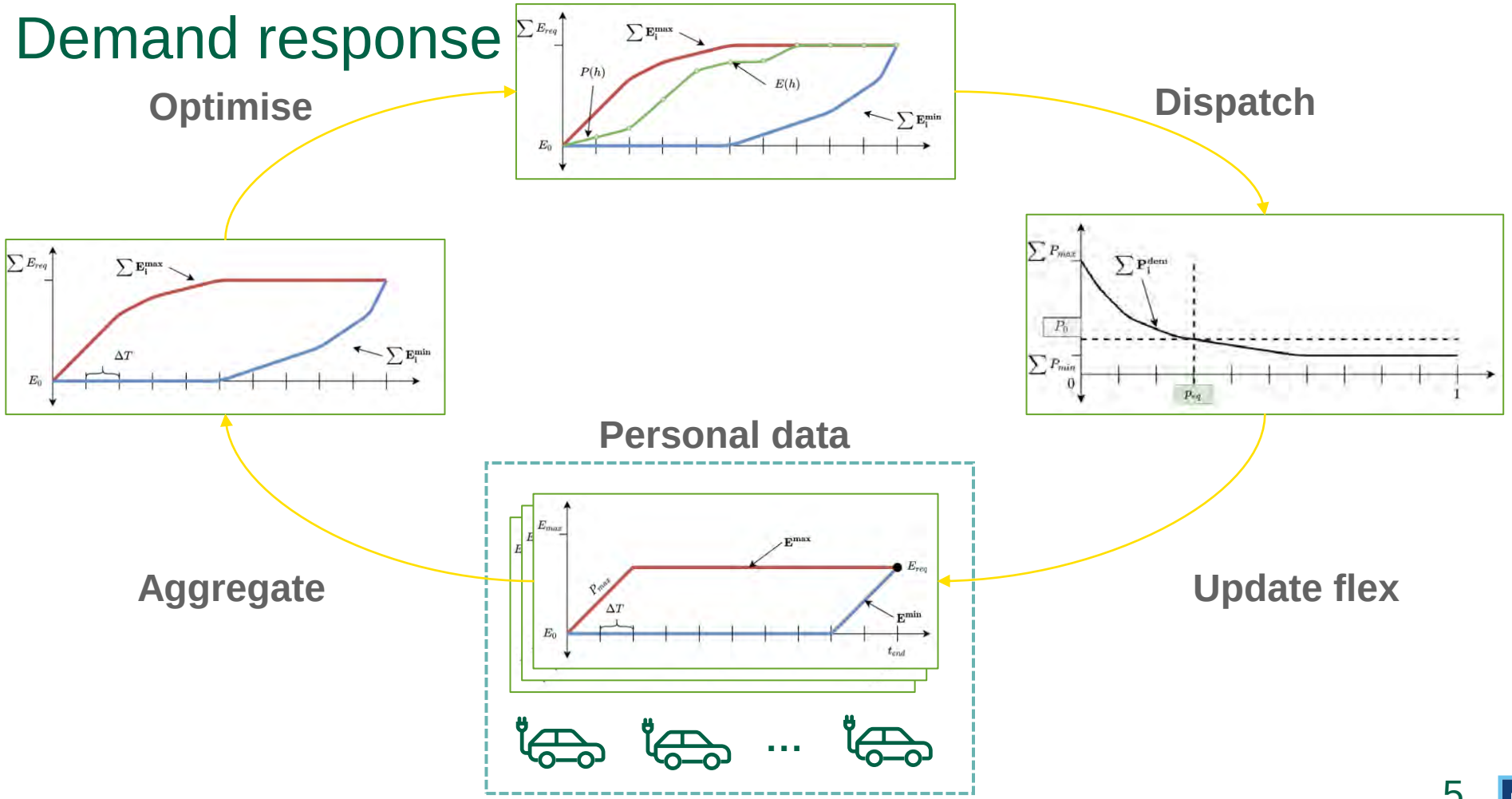


Is it that bad? – The daily picture

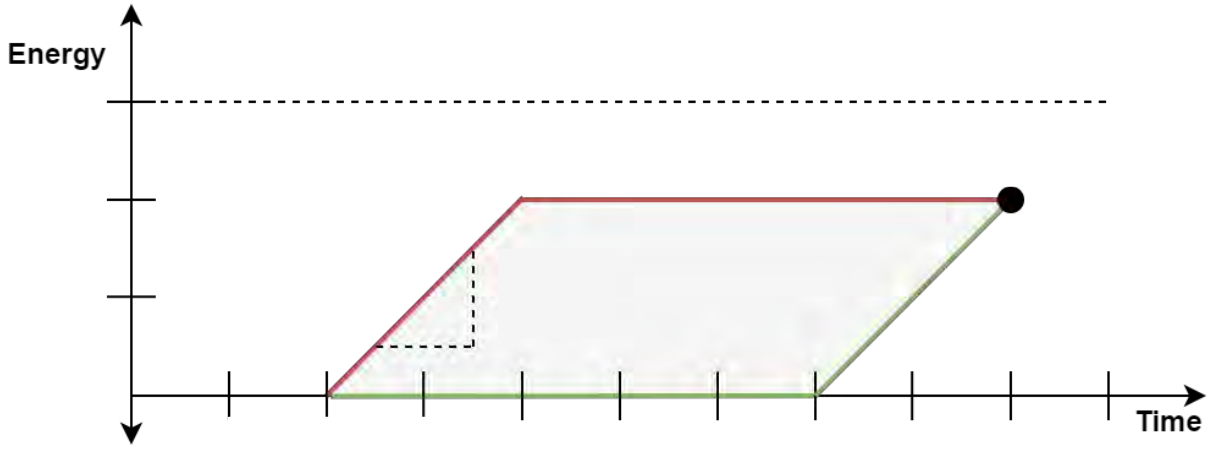


Why is privacy relevant in the smart grid?

- E.g., Demand response



What are we disclosing?



Types of appliances in the household

Technical details of the load

- Maximum capacity
- Maximum charging power

User preferences

- Available time to charge
- Departure time
- Required range

Other information

Objective : power flexibility trading from distributed resources, which is **privacy-preserving by design**

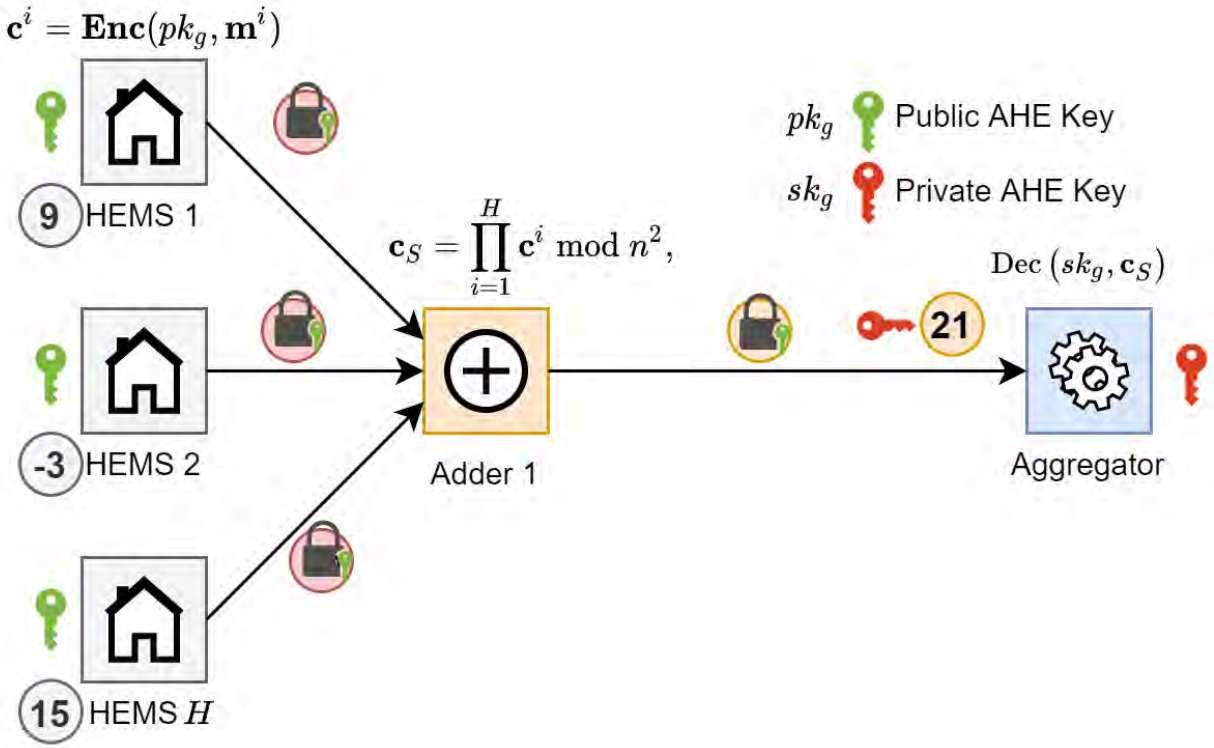
Approach to privacy

AES, RSA

Data at rest

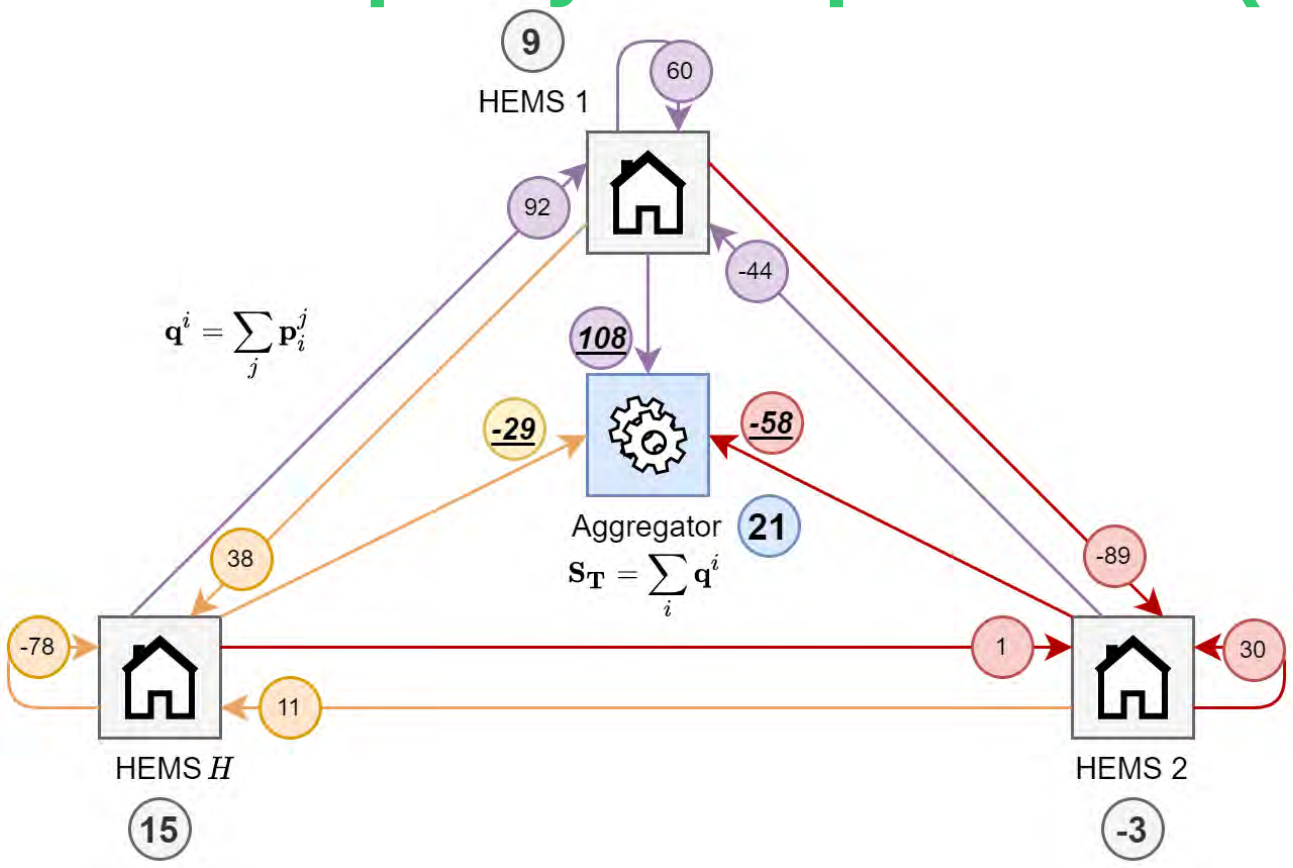
- **Computations over encrypted data** (COED)
  - keep flex data local and private
  - allowing flex trading, e.g., via an aggregator
- Main techniques:
  - Homomorphic encryption
    - ↑ Computation ↓ Communication
  - Multi-party computation
    - ↓ Computation ↑ Communication

Approach I: Additive Homomorphic Encryption (AHE)



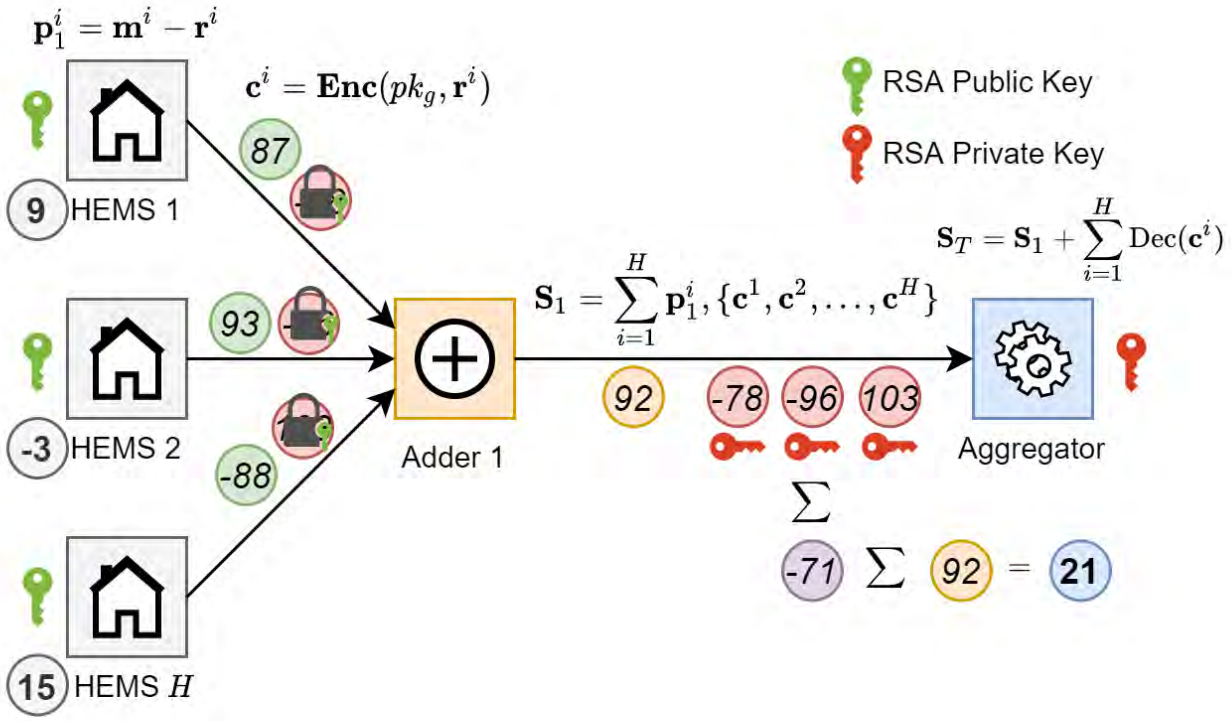
- 👍 Straightforward architecture
- 👎 Needs a 3rd party which creates collusion issues
- 👎 Computationally expensive module exponentiation and multiplication operations

Approach II: Multi-party computation (MPC)



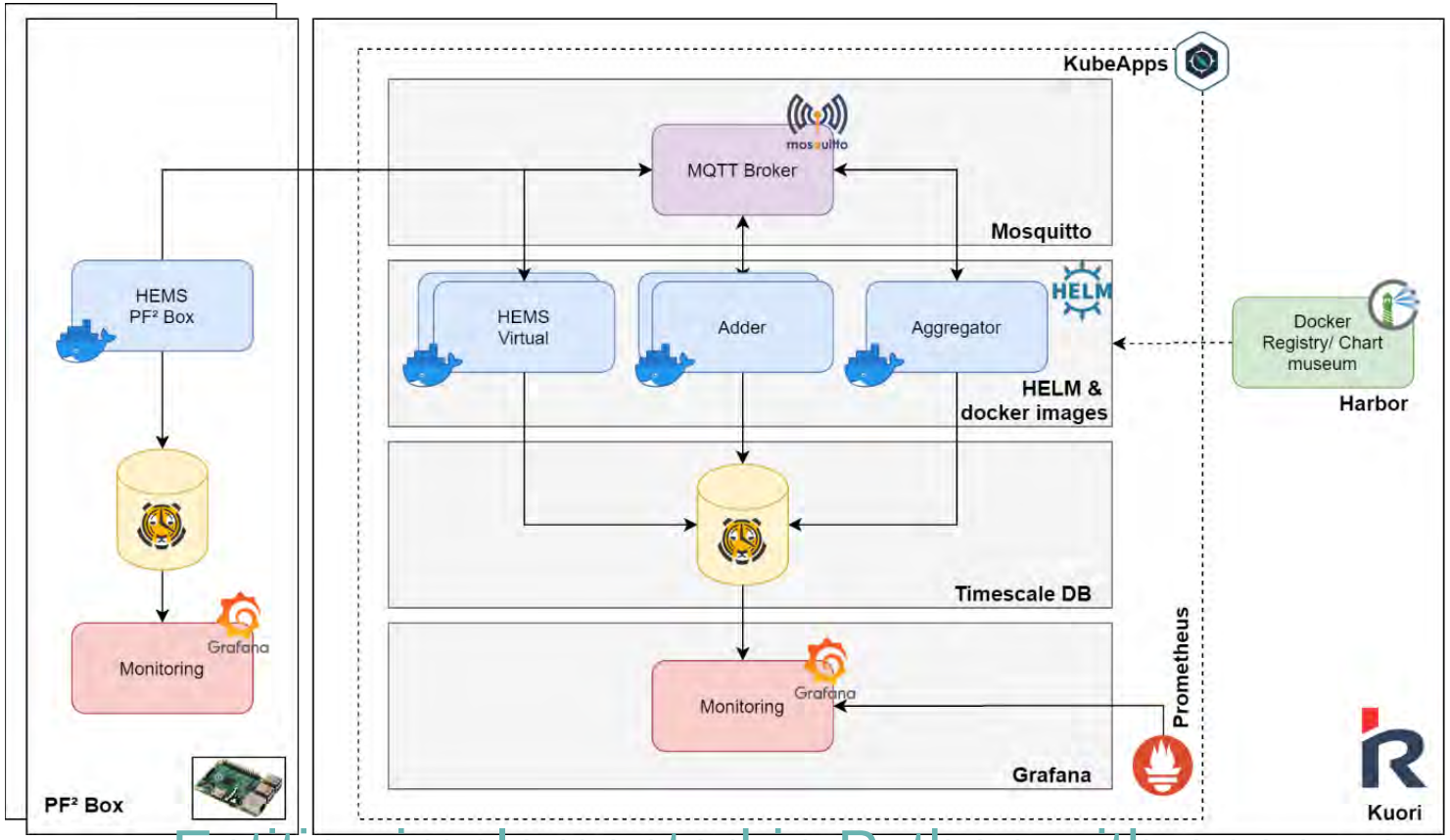
- No need for 3rd parties or complex computation
- Requires direct communication between HEMSs → Encrypt traffic
- No dropouts tolerance → Shamir's Secret Sharing

Approach III: Additive random shares + RSA



- 👍 Simple and well-established cryptographic scheme RSA (+ AES)
- 👍 Straightforward architecture + Multiple Adders
- 👎 3rd party collusion but more difficult for multiple Adders

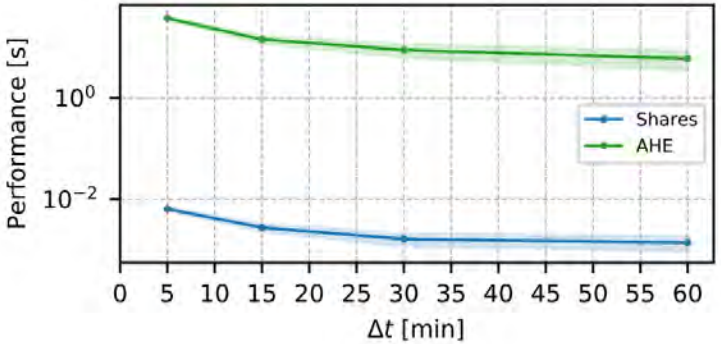
PoC deployment architecture



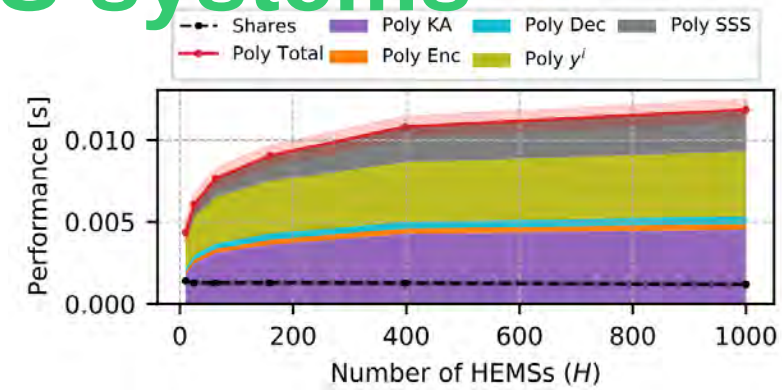
- Entities implemented in Python with open-source libraries
- Docker + Kubernetes for deployment

PoC Monitoring flexibility use

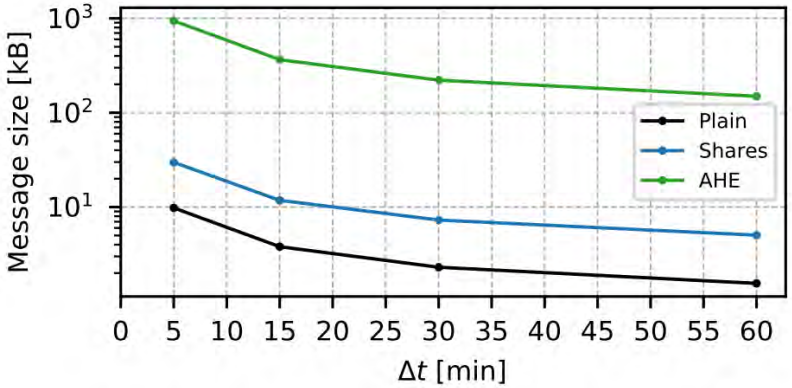
PoC Performance of HEMS systems



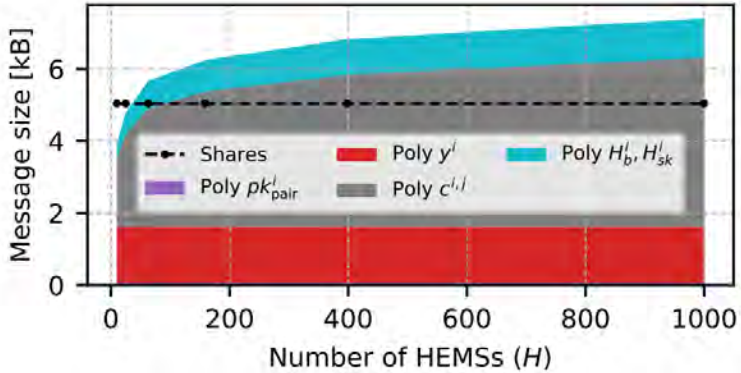
AHE vs Shares Computation



MPC vs Shares Computation



AHE vs Shares Communication



MPC vs Shares Communication

- AHE & MPC higher encryption and communication cost at HEMSs
- Communication cost for MPC depends on *t* resolution

Conclusion:
share your flexibility without sharing your data

- Privacy can be protected with very little overhead:
  - Computation: 3 ms @15 min, 6.5 ms @5 min
  - Communication: 8 kiB @15 min, 20 kiB @5 min
- Use of well-established cryptographic techniques: AES, RSA
- General COED methods are applicable but:
  - HE → high computational and communication needs on HEMS
  - MPC → high communication needs + complex scheme (dropouts)
- Flex can be managed with aggregated optimisation algorithms
  - Inter-temporal dependencies
  - Privacy-friendly optimisation

Main references

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